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Labor mobility and the level of unemployment in a currency union☆,☆☆

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ABSTRACT

Unemployment rates are substantially higher and more volatile in the euro area relative to the United States. We ask to what extent the lack of cross-country labor mobility can account for unemployment dynamics in Europe. Our analytical model incorporates downward nominal wage rigidity and an endogenous migration decision. Firms are unable to freely adjust wages during economic contractions, generating an asymmetric distribution of unemployment over the business cycle. The model is calibrated to the dynamics of unemployment and net migration in a typical euro area country. An increase in labor mobility to that observed in the United States and holding all other parameters fixed would reduce the volatility of euro area unemployment by 28% and return over 1,000,000 unemployed to the workforce. The welfare cost to a typical euro area country of the currency union is 4.1 percent of permanent consumption; increasing labor mobility reduces this cost to about 3.55 percent.

1. Introduction

Unemployment rates are persistently higher in Europe than they are in the United States. Fig. 1 plots harmonized unemployment rates in the euro area and the United States for the period 1990–2022. Except for the sharp unemployment spikes in the United States during the Great Recession and the pandemic, European unemployment rates are uniformly higher than the U.S. rate. The average unemployment rate over the 1990:01 to 2023:01 period for the euro area is 9.4 percent. Average unemployment for the United States over the 1990:01 to 2020:01 period is 6.2 percent.[1] There are many possible reasons for the generally high unemployment rates in Europe (termed "Eurosclerosis" by Giersch, 1985). Most often, the cause is attributed to European labor market rigidities caused by regulation and unionization. Such rigidities hinder workers' ability to seek alternative employment opportunities or relocate in

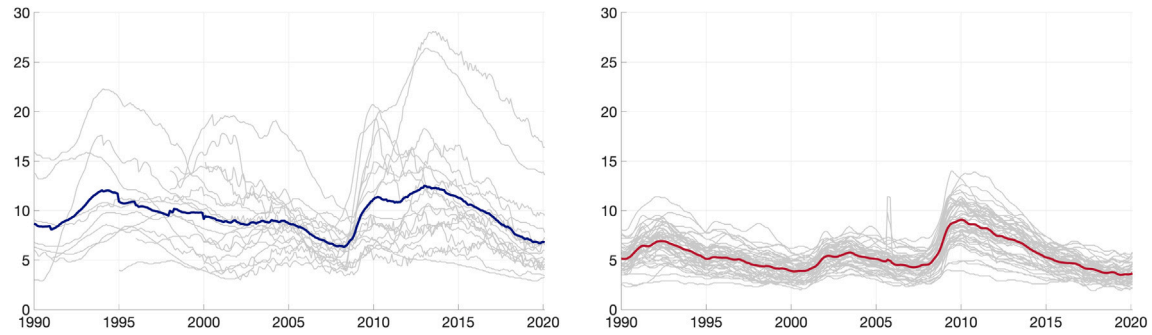
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[1] We exclude the COVID-19 period for the United States because the unprecedented, short-lived spike in the U.S. unemployment rate during the pandemic is due to lockdowns. The stable unemployment rate in the euro area during the pandemic was by design—governments implemented job retention policies that kept workers connected to firms. For a discussion, see Klitgaard (2022).

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(a) Euro area (b) United States

Fig. 1. Unemployment: Euro area vs. United States.
Note: The figure plots unemployment rates for the euro area (blue, thick line) and its member states (left), and the United States (red, thick line) and its member states. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

response to changing economic conditions. In this paper, we focus on the extent to which limited labor mobility across regions constrains labor market adjustment within a currency union.

In previous work, House et al. (2026) and Foschi et al. (2023) document higher rates of labor mobility in the United States compared to the euro area. In both currency areas, labor flows are correlated with local business cycles, with workers moving from regions with temporarily higher unemployment to regions with temporarily lower unemployment. However, the responsiveness of migration flows to local economic conditions is three times stronger in the United States than in the euro area. House et al. (2026) show that a higher level of labor mobility in the euro area would help stabilize unemployment and therefore serve as a substitute for independent monetary policy.

We return to the issue of migration in this paper, but shift the focus towards its effect on the *level* of unemployment observed in Europe rather than the effect on the cyclical variability of unemployment. While labor mobility can reduce the variance of unemployment differentials across regions, it is not clear that greater labor mobility would reduce the *average* unemployment rate. There are settings in which this happens. In particular, models that feature asymmetric business cycles with frequent shortfalls in production and employment, but relatively infrequent instances of overproduction have the property that business cycle stabilization implies an increase in average employment.

To quantify the extent to which low labor mobility in Europe results in higher average unemployment, we utilize a "plucking" model of the business cycle. The plucking model was initially suggested by Friedman (1964) and more recently advocated by Dupraz et al. (2025). Importantly, plucking models feature asymmetric cyclical variations in unemployment. In the standard business cycle framework, macroeconomic variables are assumed to fluctuate around a given trend. Friedman (1964) suggested an alternative perspective in which cycles are caused by random negative shocks that pull the economy below full employment but never cause overemployment. One might imagine a helium balloon bouncing along a ceiling. Now and then, a negative shock pulls down on the balloon's string (i.e., a recession) after which the balloon gradually floats back to the ceiling (full employment). In this setting, anything that reduces the balloon's volatility (e.g., labor mobility) will cause the average height of the balloon to rise. With zero variance, the balloon simply comes to rest at the ceiling.

Our analytical framework incorporates labor market frictions and endogenous migration in a small open economy. Following Schmitt-Grohé and Uribe (2016) (SGU), we model labor market frictions as the inability of firms to freely reduce nominal wages during economic contractions.[2] In our setup, workers consider the risk of unemployment and the anticipated wage rate in alternative locations and, subject to relocation costs, move in response to local shocks (Caliendo et al., 2019). We calibrate the model to match broad business cycle properties and the degree of labor mobility for a typical country in the euro area.

The model successfully reproduces key features of unemployment dynamics over the business cycle. It generates a positive relationship between the mean and variance of unemployment and matches the empirical correlation between net migration and unemployment in a typical euro area country. The model also captures asymmetry in unemployment rates over the cycle, generating the significant gap between the duration of expansions and contractions highlighted by Dupraz et al. (2025).

We use the model to examine the effect of increasing labor mobility in Europe. In the counterfactual, labor mobility is raised to the level observed in the United States. This increase reduces both the volatility and the mean level of unemployment, while raising aggregate output volatility and shortening the duration of expansions and contractions. In effect, increasing migration provides an additional margin of adjustment: The option to migrate eases the constraint on wage adjustment, weakening the features of the model that generate significant asymmetries. Under realistic calibrations, higher labor mobility reduces the persistence of unemployment,

[2] For recent empirical evidence on the prevalence of downward nominal wage rigidity, see e.g., Babecky et al. (2010), Caju et al. (2007), Elsby and Solon (2019), Grigsby et al. (2021), Jo (2021), Schaefer and Singleton (2022) and Hazell and Taska (2025).

cutting the half-life of an initial increase in unemployment from about 5 years to 2.5 years. This translates into lower average unemployment in euro area economies of about 0.6 percentage points.

In Section 5.2, we solve for the welfare costs of being in the currency union, under the conditions of downward wage rigidity and limited labor mobility. Considering welfare for both workers and capital owners, the average cost of being in the currency union is 4.1 percent of permanent consumption. Increased labor mobility reduces the cost to 3.6 percent of permanent consumption. Accounting for the transition from a floating to a fixed exchange rate modestly attenuates the overall welfare losses, but leaves the relative gains from increased labor mobility largely unchanged.

We also consider how migration interacts with the degree of fiscal independence. Our baseline model implicitly assumes that countries are in a fiscal union and thus can share the idiosyncratic risks of temporary gains and losses of revenue. If, in contrast, nations must balance their budget with local taxes, there is a feedback effect between labor migration and tax adjustment. A recession lowers workers' welfare and induces emigration, which in turn lowers the tax base and forces governments to raise taxes, further exacerbating the recession. This mechanism increases the responsiveness of migration to local conditions and, as a result, leads to a further reduction in average unemployment. However, it also introduces volatility in labor taxes and shifts risk onto workers, raising their welfare costs even as capital owners benefit from the stronger adjustment. This effect is reminiscent of Alessandria et al. (2020) who study this mechanism in the context of a sovereign default model. Quantitatively, the feedback effect in our model is relatively small. This is largely because our model is calibrated to European countries where migration rates and migration elasticities are relatively low.

The idea that downward nominal wage rigidity in combination with suboptimal monetary policy (such as a fixed exchange rate) can generate higher average levels of unemployment has gained traction among macroeconomists and trade economists (Schmitt-Grohé and Uribe, 2016; Dupraz et al., 2025; Rodríguez-Clare et al., 2026). Rodríguez-Clare et al. (2026) show that DNWR can rationalize the increase in unemployment in U.S. states that were particularly exposed to stronger import competition from China. Similar to House et al. (2026), their discussion of labor mobility is limited to its effect on the dispersion in unemployment. In contrast, we focus on its effects on the average level of unemployment and ask how labor mobility can mitigate the costs of suboptimal monetary policy. This complements the discussion in SGU who show that capital controls also reduce the average level of unemployment by preventing overborrowing and strong nominal wage increases during booms. More recently, Fornaro and Grosse Steffen (2024) develop a theory of fragmented monetary unions in which imperfect capital mobility generates endogenous asymmetries across regions, as capital flows amplify differences in economic conditions. Financial channels can, in principle, act as substitutes for labor mobility, as both reallocate resources in response to local shocks. However, imperfect capital mobility may itself be a source of instability, giving rise to self-fulfilling capital flows and persistent regional imbalances (Fornaro and Grosse Steffen, 2024).

2. Empirical patterns

In this section, we present evidence on unemployment dynamics across the euro area and the United States. Unemployment dynamics in both regions are characterized by asymmetries consistent with the plucking narrative. On average, countries in the euro area have both more volatile and higher unemployment rates compared to U.S. states. In both regions, countries or states with more volatile unemployment rates also have higher mean unemployment rates, and this mean–variance relationship is similar across the two regions.

2.1. Data sources

European unemployment data for individual countries are taken from the short-term labor force statistics reported by the International Labor Organization. The data are collected through the European labor force survey and are based on standard international definitions of unemployment. Unemployment rates are monthly, seasonally adjusted, and cover the years 1990 to 2023, with some countries having shorter samples.

Data for aggregate and state-level unemployment in the United States are taken from the local area unemployment statistics reported by the Bureau of Labor Statistics. All data are monthly and seasonally adjusted. The series start in 1976, and we stop our analysis at the onset of COVID in February 2020.

Data on migration in the euro area are drawn from Eurostat and are cross-checked against national sources. U.S. migration flows at the state level are based on publicly available data from the Internal Revenue Service (IRS). For detailed information on the definition and calculation of migration rates, see the discussion in House et al. (2026), and Foschi et al. (2023).

2.2. Unemployment asymmetries

Fig. 2 shows the time series of unemployment rates for individual European countries. Dark lines mark the peak of the business cycle expansion (the local minimum of the unemployment rate) and dotted lines show the onset of the recession (the local maximum of unemployment). To fix the dates of expansions and recessions, we follow the algorithm developed by Dupraz et al. (2025) and find local minima and maxima of the unemployment rates based on substantial swings in the unemployment rate (see the Appendix for details). As a reference point, we repeat the exercise for each U.S. state. The U.S. figures are provided in the appendix.

The time series on unemployment rates in Fig. 2 indicate that fluctuations in unemployment are not tightly synchronized across the euro area. The average of bilateral correlations between euro area countries is 0.49, in contrast with the cross-state correlation

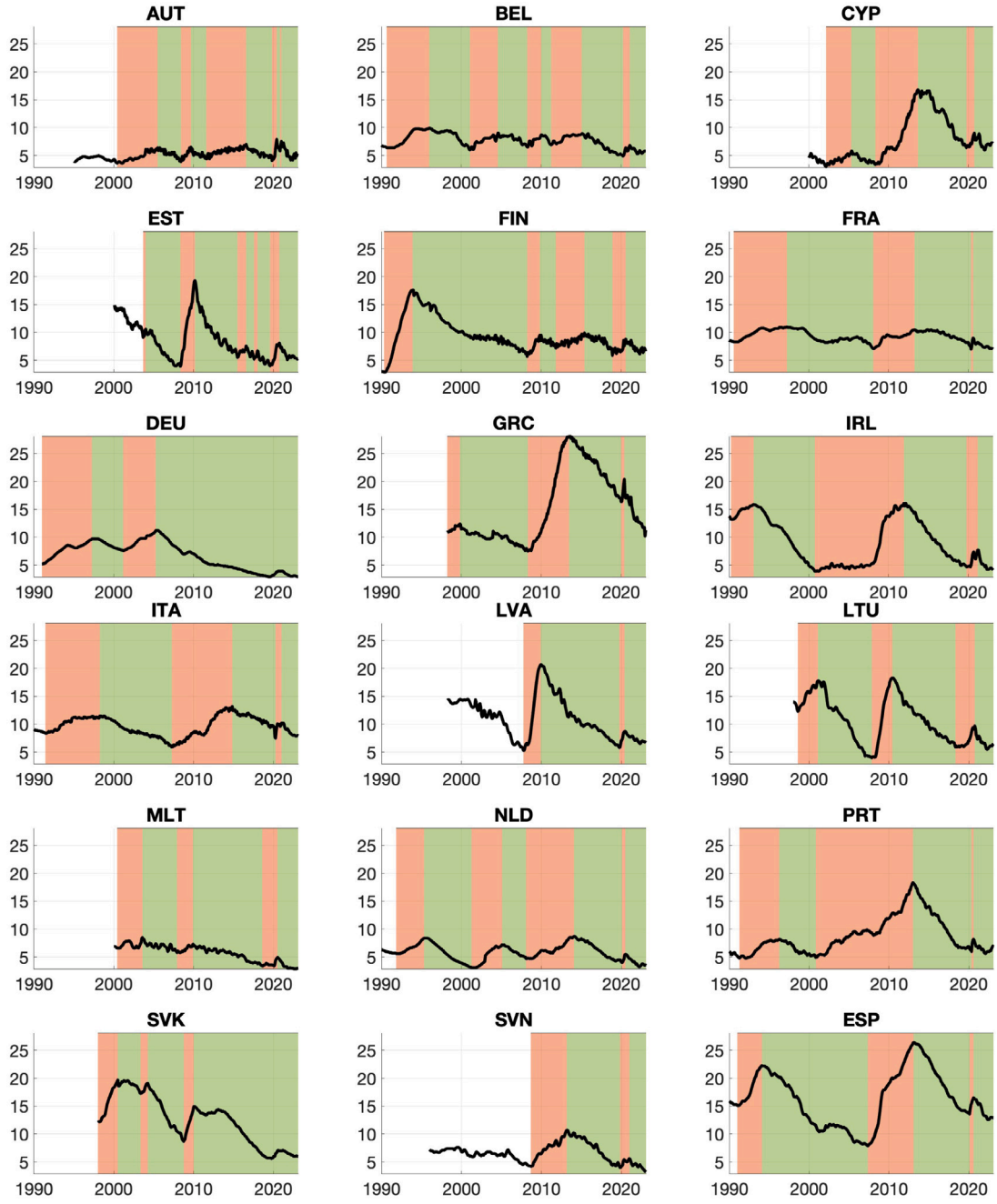


Fig. 2. Peaks and Troughs in Unemployment Rates across the Euro Area.
Notes: Figure displays unemployment rates for euro area countries. Business cycle expansions are shaded green, while business cycle contractions are shaded red. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

of 0.69. The figures are also suggestive of the asymmetry in unemployment dynamics, with smaller gaps between the end of an expansion and the next recession (i.e. the sudden increase in unemployment in the recession) compared with the longer duration of the subsequent expansion.

These features of the data are more evident in Table 1. The first two columns provide U.S. data as a point of reference, with the first column providing statistics for the United States as a whole, and the second column showing the moments found by averaging the 50 U.S. states. Similarly, column 3 provides data for the euro area aggregate, and the fourth column shows the average of the moments of the euro area member countries. Focusing on the state and country averages (columns 2 and 4), which are the relevant

Table 1
Statistics on unemployment dynamics.

	United States		Euro area	
	Aggregate	Ave state	Aggregate	Ave country
Moments				
Average unemployment rate (in %)	6.23	6.23	9.39	9.26
Std. deviation (monthly)	1.63	1.85	1.40	2.62
Skewness (monthly)	0.62	0.69	0.05	0.26
Speed				
Speed of expansions (pp/year)	0.61	0.79	0.54	1.19
Speed of contractions (pp/year)	1.18	1.37	1.39	2.63
Duration				
Duration of expansion (months)	83	81	71	75
Duration of contraction (months)	39	41	37	41
Regression				
Expansion on preceding contraction				
β	1.12	0.81	0.74	0.62
R^2	0.89	0.52	0.76	0.55
Contraction on preceding expansion				
β	−0.80	−0.21	−0.71	0.25
R^2	0.69	0.05	0.30	0.05
Time period	1976.01 - 2020.01		1990.01 - 2023.01	

Notes: Each column corresponds to a separate sample. Column 1 refers to statistics for the national U.S. unemployment rate series, column 2 refers to average statistics across U.S. state-level unemployment rate series. Columns 3 and 4 report the statistics for the euro aggregate and the average across euro area countries. **Moments**: Reports first, second and third moments of monthly unemployment rates. **Speed**: Reports the average decrease in the unemployment rate during expansions, measured in percentage points per year, and the average increase in the unemployment rate during contractions. **Duration**: Reports the duration of expansions and contractions in months. **Regressions**: The first specification ("expansion on preceding contraction") reports the coefficient in an OLS regression of the size of the subsequent expansion (percentage point fall in unemployment rate) on the size of a contraction (percentage points increase in unemployment). The second specification ("contraction on preceding expansion") reports the coefficient in an analogous regression of the size of the subsequent contraction on the size of an expansion. Regressions are pooled across U.S. states (column 2) or euro area countries (column 4).

statistics of comparison for our model, we see that both the mean and the standard deviation of unemployment are about fifty percent higher in the euro area relative to the United States.

In both regions, unemployment rates are positively skewed, meaning that most of the time, the unemployment rate is below its mean. This characteristic is somewhat more pronounced for the United States (0.69 vs. 0.26). Expansions are longer than contractions. On average, expansions in both regions take about 6.5 years, whereas contractions are only half as long. Consequently, the unemployment rate takes longer to fall during expansions than it takes to rise during contractions. In both regions, unemployment rates rise twice as quickly during contractions as they fall during expansions. On the other hand, changes in the unemployment rate in the euro area are substantially faster, as already suggested by the higher standard deviation. For instance, during contractions unemployment rates rise by 1.4 percentage points per year in the average U.S. state, whereas they rise by 2.6 percentage points in the average euro area country.

The final asymmetry reported in Table 1 tests the plucking property of the unemployment rate more directly. Following the image of a helium balloon being randomly pulled down and then gradually drifting back, the size of a contraction should be predictive about the size of the following expansion. The stronger the balloon is pulled down, the stronger the subsequent drift up. To test this, we follow Dupraz et al. (2025) and regress the amplitude (in percentage points) of an expansion on its preceding contraction. For both regions, the size of a contraction is a strong predictor for the size of the subsequent expansion with R^2 above 0.5. In the United States, the relationship is close to one-for-one: For every percentage point increase in the amplitude of a contraction, the amplitude of the subsequent expansion increases by 0.81 percentage points. In the euro area, this coefficient is somewhat smaller (0.62). More importantly, running an analogous regression of the size of a contraction on its preceding expansion has no predictive power: its R^2 is 0.05 in both regions. Hence, the size of an expansion is not predictive of the size of the following contraction.

Together, these statistics indicate a strong asymmetry in unemployment rate dynamics that motivates our model with downward nominal wage rigidity. We later evaluate our model against these moments.

2.3. Relation between the volatility of unemployment and its mean

An important correlate of the plucking view is that there should be a positive relationship between the volatility of unemployment and its average level. The stronger and more often the helium balloon gets pulled down, the more time it spends far below the ceiling. That is, countries with more volatile unemployment rates should have, on average, higher levels of unemployment rates. This prediction is borne out by the data: Fig. 3 illustrates the relationship between mean unemployment and the variation in unemployment for U.S. states (in red) and euro area countries (in blue). The slopes are in the 1.6 to 1.8 range and are statistically significantly different from zero, indicating that regions with a percentage point higher standard deviation have average

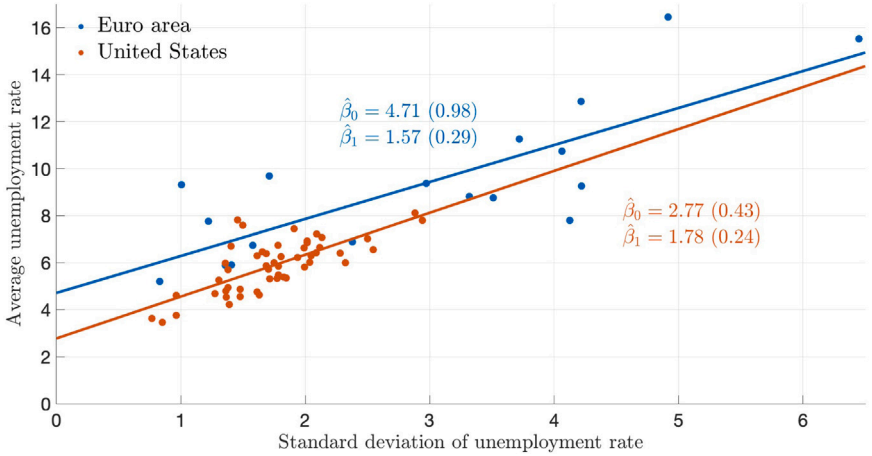


Fig. 3. Volatility and average unemployment.
Note: The figure plots the time-series standard deviation for each euro area country/U.S. state against its time-series mean. The coefficients indicate the estimated intercept (β_0) and slope (β_1) of a regression of the standard deviation on the mean. Simple standard errors are in parentheses.

unemployment rates that are about 1.7 percentage points higher. While the slopes are (both economically and statistically) similar in the two regions, the red dots of the U.S. states are bunched in the bottom left corner characterized by stable and low unemployment rates, whereas the blue dots of the euro area countries are equally spread across the regression line.

One interpretation of this figure is that stabilizing unemployment rates in the euro area could bring down their average level as well. In terms of the figure, the idea would be to move down along the regression line towards the bottom left corner. There are obviously many possible reasons why unemployment rates are more stable across U.S. states, ranging from smaller business cycle shocks to various risk-sharing mechanisms such as fiscal transfers and income diversification. In this paper, we study to what extent the lack of labor mobility in the euro area could help explain the higher volatility of unemployment rates and, consequently, their higher level.

Interpreted through the lens of the plucking model, the intercept in Fig. 3 can be described as 'structural' unemployment: It corresponds to the unemployment rate that would prevail in the absence of any fluctuations, or the height of the ceiling the balloon is bouncing along. The intercept is somewhat higher in the euro area than in the United States (4.71% vs. 2.77%), in line with the view of a more sclerotic European labor market. By subtracting the level of structural unemployment from the average unemployment rate in each region, we obtain a measure of the average cyclical unemployment rate—4.55% in the euro area and 3.13% in the United States. It is this part of the unemployment rate that stabilization policy and labor mobility can hope to bring down.

3. Model

The world economy consists of a currency union that is made up of a continuum of small open economies represented by the unit interval (Gali and Monacelli, 2008). Each economy, indexed by $i \in [0, 1]$, is of measure zero and takes the aggregate dynamics of the world economy as given. All economies are ex ante identical, but they are subject to idiosyncratic shocks that wash out at the aggregate, union-wide level. Each economy is populated by immobile capital owners and mobile workers. Capital owners accumulate capital and have access to complete financial markets. Workers are hand-to-mouth, supply labor inelastically, but can choose to migrate across economies. Each economy produces and consumes both a non-tradable and a perfectly tradable good. The stochastic driving processes are country-specific shocks to total factor productivity (TFP) in the tradable good sector. The only nominal friction is a restriction on how much nominal wages can fall in a given period. Consequently, negative TFP shocks can lead to temporary periods of unemployment while the wage gradually adjusts.

3.1. Households

The exposition focuses on a single economy belonging to the currency union, say, country i. We denote union-wide variables by an asterisk, e.g., $X_t^* \equiv \int_0^1 X_{i,t} di$. The small open economy is populated by capital owners and workers. Capital owners are immobile. Their number is given by $\mathbb{N}_i^k$. Workers are mobile and can move between the small open economy and the rest of the world if they find it optimal to do so. The number of workers in the small open economy at time t is $\mathbb{N}_{i,t-1}^w$ and the total population is[3]

$$\mathbb{N}_{i,t-1} = \mathbb{N}_i^k + \mathbb{N}_{i,t-1}^w.$$

[3] Notice the timing convention follows the convention for the capital stock. Migration takes one period, so the number of workers present in t is decided in period $t-1$.

Variables are indicated in per capita terms. For instance, $c^w_{i,t}$ is consumption of a single worker while total consumption for workers is $\mathbb{N}^w_{i,t-1}c^w_{i,t}$.

3.1.1. Capital owners

Capital owners receive utility from consumption of the final good. At any date t, their expected discounted utility is

$$\mathbb{E}_t \sum_{j=0}^{\infty} \beta^{t+j} \frac{\left(c^k_{i,t+j}\right)^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}},$$

where $c^k_{i,t}$ denotes consumption of the capital owners, σ is the intertemporal elasticity of substitution and β is the discount factor.

In any period t, capital owners rent out their capital stock, $K_{i,t-1}$, to firms at the nominal rental rate $R_{i,t}$. Capital owners also have access to state-contingent bonds that pay off in euros, the union's currency. The euros can be used to purchase units of a tradable good. The nominal exchange rate $E_{i,t}$ is the price of the home currency in euros, such that an increase in $E_{i,t}$ implies an appreciation of the home currency. In our baseline specification, we assume that country i is part of the euro currency area and thus $E_{i,t} = 1$. Capital owners also receive lump-sum transfers from the government, $T^k_{i,t}$. The capital owners use their net nominal receipts to purchase consumption goods, $c^k_{i,t}$, and investment goods, $I_{i,t}$, at price $P_{i,t}$. Capital accumulation is subject to adjustment costs (Hayashi, 1982) and follows

$$K_{i,t} = I_{i,t} + (1-\delta)K_{i,t-1} - f(I_{i,t}, K_{i,t-1}) \tag{1}$$

with $f(I,K) = \frac{\phi}{2}\left(\frac{I}{K} - \delta\right)^2$ K. The parameter $\phi \geq 0$ governs the degree of adjustment costs. As is well known, adjustment costs are necessary in small open economy settings to prevent unrealistically large capital flows driven by productivity differentials across countries.

The optimal choices of $I_{i,t}$ and $K_{i,t}$ imply a standard capital Euler equation:

$$q_{i,t} = \beta \mathbb{E}_t \left[\left(\frac{c^k_{i,t+1}}{c^k_{i,t}} \right)^{-\frac{1}{\sigma}} \left(\frac{R_{i,t+1}}{P_{i,t+1}} + q_{i,t+1}\left(1 - \delta + \frac{\phi}{2}\left(\left(\frac{I_{i,t+1}}{K_{i,t}}\right)^2 - \delta^2\right)\right)\right)\right] \tag{2}$$

where Tobin's q is $q_{i,t} = \left(1 - \phi\left(\frac{I_{i,t}}{K_{i,t-1}} - \delta\right)\right)^{-1}$. The presence of state-contingent nominal euro bonds implies that the marginal value of currency is equalized for capital owners in every country. This, in turn, implies the familiar Backus-Smith risk-sharing condition (Backus and Smith, 1993):

$$c^k_{i,t} = c^{k,*}_t \left(\frac{P^*}{E_{i,t}P_{i,t}}\right)^{\sigma}, \tag{3}$$

where $c^{k,*}_t \equiv \int_0^1 c^k_{i,t}di$ is union-wide consumption of capital owners and $P^*_t \equiv \int_0^1 P_{i,t}di$ is the union-wide price index.

3.1.2. Workers

Workers are mobile and earn only labor income. Each worker supplies L^S units of labor inelastically. Because some workers can be involuntarily unemployed in equilibrium, actual hours worked $L_{i,t}$ can be less than L^S. Workers are assumed to be hand-to-mouth and simply consume their after-tax income. As a result, their consumption is

$$P_{i,t}c^w_{i,t} = (1-\tau^w)W_{i,t}L_{i,t}, \tag{4}$$

where τ^w is a labor income tax. We introduce labor income taxes to discipline the share of aggregate consumption that goes to hand-to-mouth households.

Labor markets. Nominal wages are downwardly rigid. Following the specification in SGU, there is no constraint on wage increases, but wages can only fall by a limited amount. Specifically, the prevailing nominal wage must satisfy

$$W_{i,t} = \max\{W^{flex}_{i,t}, \eta W_{i,t-1} + (1-\eta)W^{flex}_{i,t}\} \tag{5}$$

where $W_{i,t}$ is the nominal wage, $W^{flex}_{i,t}$ is the nominal wage that would occur in the absence of any constraint, and $\eta \in [0,1]$ is a parameter that governs the rate of downward wage adjustment. If $\eta = 0$ then wages are fully flexible, $W_{i,t} = W^{flex}_{i,t}$. If $\eta = 1$ then nominal wages can never fall and $W_{i,t} = W_{i,t-1}$. The presence of wage rigidity implies that the labor market will in general not clear and some workers will be unemployed. As in Hansen (1985) and Rogerson (1988), workers agree to a risk-sharing contract that guarantees each worker has the same income at date t, though in equilibrium not all workers are employed. The unemployment rate is $ur_{i,t} = 1 - \frac{L_{i,t}}{L^S_{i,t}}$.

Migration. At the beginning of each period, workers choose to migrate or remain in their current country. Migration takes one period, and migrants work and consume in their original location.

Moving from one country to another entails a fixed moving cost, ξ, denoted in utils. Each period, workers in country i receive idiosyncratic (i.e., worker-specific) shocks for each economy j, denoted by $\epsilon_{ij,t}$. Define $v_{i,t}\left(\epsilon_{i,t}\right)$ as the value of a worker currently living in Home conditional on the aggregate state and the worker's vector of idiosyncratic shocks, $\epsilon_{i,t}$. The value of living in Home at time t is

$$v_{i,t}(\epsilon_{i,t}) = U\left(c_{i,t}^{w}\right) + \max_{j}\left\{\frac{1}{\gamma}\epsilon_{ij,t} - \xi\mathbb{1}_{i\neq j} + \beta\mathbb{E}_t\left(V_{j,t+1}\right)\right\}$$

The flow utility function $U(c)$ is $\frac{c^{1-\frac{1}{\sigma}}}{1-\frac{1}{\sigma}}$. The value $V_{j,t}$ is the expected value of $v_{j,t}\left(\epsilon_{j,t}\right)$ prior to the realization of the idiosyncratic shocks and thus, $V_{j,t}$ is the average expected utility of any worker in j at the start of time t. The parameter γ governs how strongly idiosyncratic location shocks affect migration decisions.

We follow Artuç et al. (2010) and assume that the idiosyncratic shocks are i.i.d. over time and across individuals and are distributed according to a Type-I extreme value distribution with zero mean. Given these assumptions, the expected value $V_{i,t}$ — the average utility of a worker in country i at time t — is

$$V_{i,t} = U\left(c_{i,t}^{w}\right) + \frac{1}{\gamma}\ln\left(\int_j \exp\left\{\gamma\left(-\xi\mathbb{1}_{i\neq j} + \beta\mathbb{E}_t\left(V_{j,t+1}\right)\right)\right\}dj\right). \tag{6}$$

Migration decisions depend on this average utility. The share of workers that relocate from i to j, denoted by $n_{j,t}^{i}$, is

$$n_{j,t}^{i} = \frac{\exp\left\{\gamma\left(-\xi\mathbb{1}_{i\neq j} + \beta\mathbb{E}_t\left(V_{j,t+1}\right)\right)\right\}}{\int_k \exp\left\{\gamma\left(-\xi\mathbb{1}_{i\neq k} + \beta\mathbb{E}_t\left(V_{k,t+1}\right)\right)\right\}dk}. \tag{7}$$

Naturally, an increase in expected life-time utility in country j, $\mathbb{E}_t\left(V_{j,t+1}\right)$, raises immigration to that country. The larger the value of γ, the stronger the migration response. The number of workers who move to country i at time t is

$$\mathbb{N}_{i,t}^{w} = \int_j n_{i,t}^{j}\mathbb{N}_{j,t-1}^{w}dj.$$

We can summarize the migration dynamics with two key equations. For any variable $x_{i,t}$, let $\hat{x}_{i,t}$ denote the difference between that variable and its non-stochastic steady state value; that is, $\hat{x}_{i,t} = x_{i,t} - \bar{x}_i$. Then, the expected value of being in country i relative to its non-stochastic steady state value ($\hat{V}_{i,t} = V_{i,t} - \bar{V}_i$) satisfies

$$\exp\left\{\gamma\left(\hat{V}_{i,t} - \hat{U}_{i,t} - \beta\mathbb{E}_t\hat{V}_{t+1}^{*}\right)\right\} = (1-\bar{n}_i)\exp\left\{\gamma\beta\mathbb{E}_t\left(\hat{V}_{i,t+1} - \hat{V}_{t+1}^{*}\right)\right\} + \bar{n}_i, \tag{8}$$

where $U_{i,t} = U\left(c_{i,t}^{w}\right)$, $V_t^{*} = \int V_{j,t}dj$, and the population of workers satisfies

$$\mathbb{N}_{i,t}^{w} = \mathbb{N}_{i,t-1}^{w}\left(1 - \exp\left\{-\gamma\left(\hat{V}_{i,t} - \hat{U}_{i,t} - \beta\mathbb{E}_t\hat{V}_{t+1}^{*}\right)\right\}\bar{n}_i\right) + \exp\left\{\gamma\beta\mathbb{E}_t\left(\hat{V}_{i,t+1} - \hat{V}_{t+1}^{*}\right)\right\}\bar{n}_i\bar{N}_i^{w}. \tag{9}$$

3.2. Firms

Each economy produces intermediate goods $Y_{i,t}$ from labor and capital. The intermediate goods are then transformed into either a non-tradable good or a perfectly tradable good. The transformation into tradables is subject to productivity shocks. In a final stage, non-tradables and tradables are aggregated into a final good $Q_{i,t}$.

Intermediate goods. Perfectly competitive producers use capital and labor to produce an intermediate good $Y_{i,t}$ that they sell domestically at price $P_{i,t}^{Y}$. They choose inputs to maximize profits

$$P_{i,t}^{Y}\mathbb{N}_{i,t-1}Y_{i,t} - W_{i,t}\mathbb{N}_{i,t-1}^{w}L_{i,t} - R_{i,t}\mathbb{N}_i^{k}K_{i,t-1}$$

subject to the production function

$$\mathbb{N}_{i,t-1}Y_{i,t} = \left(\mathbb{N}_i^{k}K_{i,t-1}\right)^{\alpha_K}\left(\mathbb{N}_{i,t-1}^{w}L_{i,t}\right)^{\alpha_L}. \tag{10}$$

By assuming $\alpha_K + \alpha_L < 1$, we implicitly introduce a fixed local factor — owned by capital owners — which we interpret broadly as land and other immobile inputs. The optimal choice of inputs implies standard demand curves for labor and capital. Namely,

$$W_{i,t} = \alpha_L P_{i,t}^{Y}\left(\mathbb{N}_i^{k}K_{i,t-1}\right)^{\alpha_K}\left(\mathbb{N}_{i,t-1}^{w}L_{i,t}\right)^{\alpha_L-1} \tag{11}$$

and

$$R_{i,t} = \alpha_K P_{i,t}^{Y}\left(\mathbb{N}_i^{k}K_{i,t-1}\right)^{\alpha_K-1}\left(\mathbb{N}_{i,t-1}^{w}L_{i,t}\right)^{\alpha_L}. \tag{12}$$

Non-tradables and tradables. Another set of competitive firms transforms the intermediate good into non-traded goods, $Y_{i,t}^{N}$, or tradable goods, $Y_{i,t}^{T}$, using linear production functions. The tradable-good sector experiences productivity shocks denoted by $A_{i,t}$. The price of the non-traded good is then $P_{i,t}^{N} = P_{i,t}^{Y}$ and the price of the tradable good, which is common for all countries, is $P^{T} = \frac{P_{i,t}^{Y}}{A_{i,t}}$.

Productivity $A_{i,t}$ evolves according to a finite state Markov chain with transition matrix Λ ($\Lambda(a',a)$ is the probability that $A_{i,t+1} = a'$ conditional on $A_{i,t} = a$). This process is the same for all the countries in the union.

Final goods. Finally, a third group of competitive firms combines the non-traded goods and the tradable goods into a single final good, which is used for domestic consumption and investment. The production of the final good, $Q_{i,t}$, is given by a standard Cobb–Douglas aggregator:

$$Q_{i,t} = \left(Q_{i,t}^N\right)^{\omega} \left(Q_{i,t}^T\right)^{1-\omega}.$$

where $Q_{i,t}^N$ and $Q_{i,t}^T$ are the inputs of non-tradable and tradable goods, and ω is the weight on non-tradables. The optimal choice of tradable and non-tradable inputs requires

$$P_{i,t}^N Q_{i,t}^N = \omega P_{i,t} Q_{i,t} \tag{13}$$

$$P^T Q_{i,t}^T = (1-\omega) P_{i,t} Q_{i,t}. \tag{14}$$

which implies that the nominal price of the final good is

$$P_{i,t} = \left(\frac{P_{i,t}^N}{\omega}\right)^{\omega} \left(\frac{P^T}{1-\omega}\right)^{1-\omega}. \tag{15}$$

3.3. Government

The government collects taxes on labor income at a constant rate τ^w to finance a lump sum transfer to capital owners, $T_{i,t}^k$. The transfer to the capital owner is adjusted such that the government maintains a balanced budget:

$$\mathbb{N}_i^k T_{i,t}^k + \tau^w \mathbb{N}_{i,t-1}^w W_{i,t} L_{i,t} = 0. \tag{16}$$

This formulation implies that fluctuations in tax revenues, driven by changes in labor income, are absorbed entirely by capital owners through variations in the transfer. Because capital owners have access to financial markets, they can smooth these fluctuations.

The burden of fiscal adjustment does not fall on workers in the baseline economy. In Section 5.5, we relax this assumption by allowing the labor tax rate to adjust over the business cycle, introducing an additional margin through which fiscal adjustment can be shared between workers and capital owners.

3.4. Market clearing conditions

There are two sets of market clearing conditions. The first set deals with market clearing in each individual small open economy. The second set reflects the market clearing conditions for the currency union as a whole.

For each economy i, the markets for intermediate goods, nontradable goods, and final goods all clear in equilibrium.

$$Y_{i,t} = Y_{i,t}^N + \frac{Y_{i,t}^T}{A_{i,t}}$$

$$Y_{i,t}^N = Q_{i,t}^N$$

and

$$\mathbb{N}_{i,t-1} Q_{i,t} = \mathbb{N}_i^k c_{i,t}^k + \mathbb{N}_{i,t-1}^w c_{i,t}^w + \mathbb{N}_i^k I_{i,t}. \tag{17}$$

For the currency union as a whole, the market clearing condition for tradable goods requires

$$\int_0^1 \mathbb{N}_{i,t-1} Y_{i,t}^T di = \int_0^1 \mathbb{N}_{i,t-1} Q_{i,t}^T di.$$

The sum of all the countries' populations of workers must equal the fixed total population of workers $\mathbb{N}^{w,*}$,

$$\int_0^1 \mathbb{N}_{i,t}^w di = \mathbb{N}^{w,*},$$

Finally, for later reference, we note that per-capita real GDP for any single country is defined as total expenditure on domestically produced goods, evaluated at steady-state prices. That is,

$$GDP_{i,t} = \bar{P}_i Q_{i,t} + \bar{P}^T \left(Y_{i,t}^T - Q_{i,t}^T\right).$$

and aggregate consumption in country i is simply the sum of capital owners' consumption and workers' consumption:

$$\mathbb{N}_{i,t-1} C_{i,t} = \mathbb{N}_i^k c_{i,t}^k + \mathbb{N}_{i,t-1}^w c_{i,t}^w.$$

3.5. Equilibrium

We focus on the behavior of a single economy that is a part of the currency union. Because each country is assumed to be sufficiently small (measure zero), country-specific outcomes do not affect the currency union, which we take to be in a stationary equilibrium.

We use an extension of the standard recursive competitive equilibrium (RCE) concept, modified to embed the small open economy in a currency union that is itself in a stationary equilibrium. For a single small open economy, the equilibrium can be described by four state variables.[4] There is one exogenous state variable, TFP (A), and there are three endogenous state variables, the capital stock (K_{-1}), the population of workers $\mathbb{N}^w_{-1}$, and last period's nominal wage (W_{-1}). Let the state of the small open economy be denoted by $S = \{A, K_{-1}, \mathbb{N}^w_{-1}, W_{-1}\}$. We now have the following definition:

Definition. A Stationary Embedded Recursive Competitive Equilibrium is a set of household allocation functions $c^k(S)$, $c^w(S)$, $\mathbb{N}^w(S)$, $I(S)$ and $K(S)$; a location value $V(S)$; an employment allocation $L(S)$; nominal prices $R(S)$, $W(S)$, $W^{flex}(S)$, $P^Y(S)$ and $P(S)$; a value $V^* \in \mathbb{R}$ and a distribution $\pi(S)$, such that:

1. The price of the final good is $P(S) = P^T A^\omega \left[\omega^\omega (1-\omega)^{1-\omega}\right]^{-1}$, the price of the intermediate good is $P^Y(S) = P^T A$, the flexible wage is $W^{flex}(S) = \alpha_L A P^T (\mathbb{N}^k K_{-1})^{\alpha_K} (\mathbb{N}^w_{-1} L^S)^{\alpha_L - 1}$ and $W(S)$ satisfies the downward nominal rigidity condition (5).
2. The household allocation functions $c^k(S)$, $c^w(S)$, $\mathbb{N}^w(S)$, $I(S)$ and $K(S)$ are optimal (i.e., they satisfy (1), (2), (3), (4) and (9)).
3. The firms maximize profits (i.e., $L(S)$ and $K(S)$ satisfy (11) and (12)).
4. The location value $V(S)$ satisfies (8).
5. The union-wide population is constant,

$$\int_S \mathbb{N}^w(S)\pi(S)dS = \mathbb{N}^{w,*},$$

and the market clearing conditions for tradables, non-tradables, intermediates and final goods together are satisfied such that

$$\int_S \mathbb{N}(S)P^Y(S)Y(S)\pi(S)dS = \int_S P(S)\left[\mathbb{N}^k(c^k(S) + I(S)) + \mathbb{N}^w(S)c^w(S)\right]\pi(S)dS,$$

with $\mathbb{N}(S) = \mathbb{N}^w(S) + \mathbb{N}^k$.

6. The distribution π is implied by $W(S)$, $K(S)$ and $\mathbb{N}^w(S)$: for all (a, w, k, N) we have

$$\pi(a, w, k, N) = \sum_A \iiint \Lambda(a, A)\pi(A, W_{-1}, K_{-1}, \mathbb{N}^w_{-1})\mathbb{I}_S(w, k, N)\, dW_{-1}\, dK_{-1}\, d\mathbb{N}^w_{-1}.$$

where $\mathbb{I}_S(w, k, N) = 1$ if $W(S) = w$, $K(S) = k$, and $\mathbb{N}^w(S) = N$ and 0 otherwise.

4. Model solution and parameter values

4.1. Model solution

Because the model embodies non-linear, asymmetric equilibrium dynamics, we numerically solve the model using a global solution method. Specifically, we adapt the FiPIt algorithm for solving models with occasionally binding constraints in Mendoza and Villalvazo (2020) to our setting. We obtain a solution for a finite number of grid points in the model state space. The state space used in the numerical solution is the same as the one used in the equilibrium definition: $S = \{A, K_{-1}, \mathbb{N}^w_{-1}, W_{-1}\}$. The numerical state space includes 5 productivity levels, 20 wage levels, 15 population levels, and 30 capital stock levels. The Markov chain is constructed using the Rouwenhorst method to approximate a continuous AR(1) process in logs, where ρ_A governs the persistence of productivity and σ_A the standard deviation of innovations. Once we have computed equilibrium reactions on this grid mesh, we compute reactions for intermediate points by linear interpolation.

We embed this algorithm into an outer loop to solve an additional fixed-point problem. We adjust the average expected utility a worker gets abroad, V^*, to ensure that population movements balance at the union level and satisfy .[5] To calculate the average values for net migration (and all other moments), we simulate the model 10,000 times for 30 years (plus an initial 125-year burn-in period) and then average the moments across the 10,000 series. Details of our solution method are included in the appendix.

[4] Since this could be any country in the currency union, we suppress the subscript i.

[5] Technically, we also need to adjust the union-wide consumption of capital owners, $c^{k,*}_t$, to ensure that goods markets clear at the union level (17). But in practice, $c^{k,*}_t$ does not affect the optimal choices for $W_{i,t}$, $\mathbb{N}^w_{i,t}$ and $K_{i,t}$. Thus, we can back out the value for $c^{k,*}_t$ from (17) after we have solved for the optimal choices.

Table 2
Calibration.

Parameter		Value	Target
Calibrated parameters			
Discount factor per quarter	β	0.995	2% annual real interest rate
Coefficient of risk aversion	$\frac{1}{\sigma}$	1.5	Standard value
Labor share	α_L	0.62	62% labor share, European Commission (DG ECFIN) (2019)
Capital share	α_K	0.28	Assumes 10% share of land Hsieh and Moretti (2019)
Labor tax rate	τ^w	0.33	62% worker consumption share, see text
Share non traded goods	ω	0.67	33% import share, TiVA (2005–2015)
Depreciation rate per quarter	δ	0.025	Standard value
Migration cost	ξ	$4.82/\gamma$	0.75% annual migration rate (HPT)
Estimated parameters			
Capital adjustment cost	ϕ	1.16	Std dev of I to GDP: 2.78
Migration elasticity	γ	0.24	Net migration response to unempl differentials (HPT)
Wage rigidity per quarter	η	0.997	Std dev of unemployment rate: 2.625%
Shock persistence per quarter	ρ_A	0.97	Duration of expansions: 75 months
Shock volatility	σ_A	0.0023	Average cyclical unemployment rate: 4.55%

4.2. Parameter values

The model is expressed at a monthly frequency and is calibrated to match an average euro area country. We partition the parameters into a set of calibrated parameters and a set of estimated parameters. Parameters that have commonly accepted values used in the international business cycle literature, or parameters that have direct analogs in the data (e.g., the labor share) are calibrated accordingly. Taking the calibrated parameters as given, we estimate the remaining parameters. Table 2 lists the parameter values for the benchmark case.

Calibrated parameters. We set the discount factor to match an annual real interest rate of 2 percent. The intertemporal elasticity of substitution is set to $\sigma = \frac{2}{3}$. We set $\alpha_L = 0.62$, in line with a labor share of 62 percent (see e.g. European Commission (DG ECFIN), 2019). We set the share of the fixed local factor to 10 percent, consistent with standard macro calibrations that assign the residual factor share to non-reproducible inputs (Hsieh and Moretti, 2019). This implies $\alpha_K = 0.28$.

The quarterly depreciation rate is set to the standard value of $\delta = 0.025$. This implies an investment-to-GDP ratio of 23%, slightly above the 21% observed in the euro area. To calibrate the share of tradables, $1 - \omega$, we note that all tradables are imported (due to our SOE assumption). We therefore use data on the share of countries' final demand that stems from imports to calibrate $1 - \omega$. According to the Trade in Value Added database, the average import share of euro area countries between 2005 and 2015 was 33%. We set τ^w such that the share of consumption going to workers in the non-stochastic steady state is α_L. This implies a labor tax rate of 23%.[6]

The migration cost parameter ξ can be derived from the observed migration rate. In the non-stochastic steady state, the number of workers that stay relative to the number of workers that leave, $\frac{\bar{n}_i^i}{1-\bar{n}_i^i}$, is equal to $\exp(\gamma\xi)$.[7] Observing a large number of stayers therefore indicates a high migration cost (given γ). We use this expression to directly condition on the observed migration rates rather than explicitly solving for ξ. We target a monthly migration rate that is consistent with the annual figure of 0.75 percent reported by House et al. (2026) for euro area countries over 1995–2018.[8]

Estimated parameters. We estimate the standard deviation of the idiosyncratic location preference shocks ($\frac{1}{\gamma}$), the capital adjustment cost parameter (ϕ), the wage rigidity parameter (η), as well as the shock persistence and shock volatility (ρ_A and σ_A). Our parameter estimates are chosen to match the following five moments in the simulated data: (i) the OLS slope coefficient of net migration on unemployment, (ii) the volatility (standard deviation) of investment relative to the volatility of GDP, (iii) the average cyclical unemployment rate,(iv) the duration of expansions, and (v) the volatility of the unemployment rate.

While all parameters are estimated simultaneously, parameters are tightly linked to specific moments. The parameter γ governs the sensitivity of migration rates to utility differentials. Higher values of γ make migration more responsive to fluctuations in workers' expected labor income. House et al. (2026) report that for the sample of euro area countries, an increase in a country's

[6] Whereas estimates of the labor income share exist, the share of consumption financed by labor income is not easily measured. On the one hand, labor income is more heavily taxed than capital income—effective tax rates for labor vary around 40% for labor and 20% for capital (see e.g. Carey and Tchilinguirian, 2000); on the other hand, households receiving capital income have higher saving rates. We implicitly assume that these two effects cancel out such that the split between workers and capital owners for consumption is 62-38.

[7] See the Appendix for a full derivation.

[8] Given our estimate of γ and the assumed utility function, we can translate ξ (expressed in utils) into consumption-equivalent terms. Specifically, we ask what permanent consumption differential between two countries would compensate for the one-time cost of moving. We find that a worker would be indifferent between remaining in the poorer country and relocating to the richer one if the consumption gap is approximately 5.3% (see the Appendix for details). While this number seems high relative to actual moving costs, the parameter ξ also captures indirect and non-pecuniary costs such as separation from extended family and friends, cultural and language barriers, higher costs of integrating into the labor market, institutional barriers, etc.

Table 3
Model and data comparison.

	Data	Model
Targeted moments		
Mean cyclical unempl. (%)	4.55	4.55
SD unempl. (%)	2.62	2.63
Reg. net migr. on unempl.: β	−0.08	−0.08
SD I/SD GDP	2.78	2.77
Duration of expansion (months)	75	75
Business Cycle moments		
SD GDP (%)	3.04	3.11
SD C/SD GDP	0.94	0.39
SD NX/SD GDP	0.81	1.08
SD net. migr. (%)	0.27	0.23
Corr(C, GDP)	0.88	0.84
Corr(unempl.,GDP)	−0.41	−0.92
Corr(NX,GDP)	−0.27	0.46
Corr(net migr., GDP)	0.28	0.65
Pers. of GDP	0.91	0.95
Pers. of C	0.91	0.95
Pers. of I	0.87	0.86
Pers. of unempl.	0.96	0.94
Plucking statistics		
Duration of expansion (months)	75	75
Duration of contraction (months)	41	54
Speed of expansion (pp/yr)	1.2	3.6
Speed of contraction (pp/yr)	2.6	4.6
Reg. expansion of prev. contraction: R^2	0.55	0.14
Reg. contraction of prev. expansion: R^2	0.05	0.01

Notes: Statistics calculated at quarterly frequency unless noted otherwise. Time-series for GDP, consumption and investment are transformed into logs and then linearly detrended. Model statistics are averages from simulations of 10,000 countries.

unemployment rate by one percentage point is associated with a net outflow of 0.08 people, as suggested by the slope coefficient of an OLS regression of net migration on unemployment. The capital adjustment cost ϕ strongly influences the volatility of investment. Absent any adjustment costs, investment would be extremely volatile because capital owners arbitrage any TFP differentials across countries. To match the volatility observed in the data, we need to set $\phi = 1.16$. The shock volatility and the wage rigidity parameter influence the volatility of unemployment and its mean. With $\eta = 0$, wages are completely flexible, and the model would not generate any unemployment. As η is increased, the average level of unemployment rises, for a given level of unemployment volatility. By simultaneously adjusting η and σ_A we can match the volatility of unemployment and its mean. This results in a wage rigidity of $\eta = 0.997$.[9] The estimated shock volatility of 0.23% is fairly low and indicates that the model is able to amplify shocks. Finally, a shock persistence of 0.968 per quarter allows us to match the duration of 75 months for the average expansion.

4.3. Model and data comparison

Table 3 shows key moments of the baseline model together with data from euro area countries. The first set of moments refers to targeted moments, such as average cyclical unemployment and the volatility of unemployment. Since we have as many moments as free parameters, we are able to match those moments quite well.

The second set of moments is standard business cycle moments. Even though our model is fairly simple and abstracts from several elements that are common to medium-size DSGE models, the model performs surprisingly well. The model captures the volatility of per-capita real GDP (3 percent in the data and 3.1 percent in the model), the procyclical behavior of consumption, investment, and net migration, as well as countercyclical unemployment. Persistence of all macro variables is similarly high in both data and model. The main shortcomings are a path of consumption that is somewhat too smooth compared to the data, a common finding in the international RBC literature (Backus et al., 1992), and net exports that are procyclical rather than countercyclical.

The model performs reasonably well in generating asymmetries in unemployment rates. However, the effects are somewhat stronger in the data: Contractions are substantially shorter than expansions, lasting 41 months in the data and 54 months in the model (compared to 75 months for expansions in both data and model). This asymmetry also shows up in the speed of contractions. The unemployment rate rises faster during recessions than it falls during expansions, consistent with the data.

[9] To understand this value, note that the extent to which wages can fall depends on the level of unemployment: If the unemployment rate is 10%, then our choice for α_L implies that wages are by about 4% too high. Then, wages can fall by about $12 \times (1 - \eta) \times 4\% = 0.192\%$ per year. This is a substantially higher value for η than what is used in SGU, Born and Pfeifer (2020) and Rodríguez-Clare et al. (2026), who let nominal wages fall by a maximum of 2% a year.



Fig. 4. Impulse response to TFP shock.
Note: The figure plots the impulse response to a positive (green) and a negative (red) TFP shock based on model simulations. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

This is also illustrated in Fig. 4, which displays the impulse responses for both positive and negative TFP shocks derived from the model simulation. The impulse responses depend on the state in the period the shock occurs; the figure reflects the average across simulations and states. Negative TFP shocks lead to a stronger unemployment response and a weaker wage response. A drop in TFP by about 1.4 percent increases unemployment by more than 3.5 percent over the first year, whereas an equally sized increase in TFP reduces unemployment by less than 3 percent. Instead, it increases wages by about 0.4 percent, whereas wages fall by less than 0.1 percent over the first year in response to a negative TFP shock.

5. Effects of labor mobility in a currency union

We now consider several counterfactuals to understand the role of labor mobility in this model. First, we re-run the model assuming each economy can conduct its own monetary policy. By comparing the baseline model to this alternative monetary policy setup, we can pin down the 'costs' of the currency union. These costs necessarily depend on the alternative monetary policy, and we assume that countries would implement the optimal monetary policy. This generates the maximum gap between the baseline model and the alternative scenario. As shown in SGU, countries could completely undo the inefficiencies arising from the DNWR constraint by adjusting their exchange rate (i.e., by depreciating their currency in response to a negative TFP shock). Hence, the allocations obtained in this counterfactual correspond to the allocations obtained under flexible wages.

Second, we consider a case with DNWR and higher labor mobility. Specifically, we adjust γ, the inverse of the standard deviation of the idiosyncratic preference shocks in the workers' utility function (6) to match the slope coefficient observed in U.S. data: As reported in House et al. (2026), while the slope coefficient of regressing net migration on unemployment is −0.08 for euro area countries, it is about −0.26 for U.S. states. To match this slope coefficient, we need to raise γ from 0.24 to 1.74. As in the case with baseline labor mobility, we consider two types of monetary policy under this high-mobility scenario: a fixed exchange rate and a floating exchange rate with optimal monetary policy. By comparing these two monetary policy scenarios under both baseline mobility and high mobility, we can then study how labor mobility affects the costs of the currency union.

5.1. Effects on macroeconomic aggregates

Table 4 displays moments for the four scenarios: the baseline scenario (column 1), the optimal monetary policy scenario (column 2), the high mobility scenario (column 3) and the high-mobility + optimal monetary policy scenario (column 4). For each scenario, we re-run the model and calculate averages based on the simulated data. The table ignores any transition dynamics of switching from one policy to the other and instead reports average welfare statistics in the new steady state.

Panel A reports the standard deviations of key macro variables under these different scenarios. Optimal monetary policy helps stabilize per-capita GDP and consumption, as well as the unemployment rate. The volatility of GDP falls from roughly 3% to 1% while the fluctuations in per capita consumption and unemployment are nearly eliminated. In the high labor mobility case, optimal monetary policy also reduces fluctuations though the magnitudes of the reduction are somewhat smaller.

Panel B reports the means of the macro variables, expressed as the percent deviation from the non-stochastic steady state. In the baseline scenario, the average unemployment rate is 4.55%. Accompanying the lower labor input, the average capital stock and average GDP are also below their non-stochastic steady state levels. Similarly, consumption for both workers and capital owners is about 4% lower than in the non-stochastic steady state.

Table 4
Policy experiments.

	Baseline mobility		High mobility	
	Fixed	Float	Fixed	Float
Panel A: Standard deviations (%)				
GDP	3.11	1.06	2.67	1.16
Unemployment	2.62	0.00	1.80	0.00
Population	0.95	0.09	1.68	0.48
Consumption	1.22	0.06	0.68	0.12
Cons. workers	2.21	0.27	1.32	0.22
Cons. capital owners	0.52	0.52	0.53	0.52
Panel B: Avg. % dev. from non-stochastic steady state				
GDP	−4.14	−0.02	−3.63	−0.02
Unemployment	4.55	0.00	3.97	0.00
Capital stock	−4.27	0.01	−3.82	0.00
Consumption	−4.00	0.00	−3.47	0.01
Cons. workers	−4.10	−0.00	−3.56	−0.00
Cons. capital owners	−3.89	0.01	−3.35	0.02
Panel C: Welfare costs of currency union				
Cost for worker	4.19		3.63	
Cost for capital owner	3.97		3.43	
Average cost	4.11		3.55	
Panel D: Welfare costs including transition				
Cost for worker	3.80		3.44	
Cost for capital owner	3.51		3.14	
Average cost	3.69		3.33	

Notes: Statistics are calculated at quarterly frequency. GDP and consumption are calculated in per-capita terms. In panel C, welfare costs of the currency union are calculated by comparing stationary equilibria with and without a peg. In panel D, the calculation of welfare costs of the currency union also take into account the transition from a regime with floating exchange rates to a regime with fixed exchange rates.

Under optimal monetary policy, the capital stock is slightly above the non-stochastic steady state level. Optimal monetary policy eliminates unemployment, but as seen in Panel A, it does not eliminate the volatility of consumption for capital owners. Due to the presence of non-traded goods, access to complete financial markets does not fully insure capital owners against country-specific TFP fluctuations. As a result, capital owners engage in precautionary saving, which raises the average capital stock above its non-stochastic steady-state level.

Despite the higher average level of the capital stock, GDP is slightly below its non-stochastic steady-state level due to decreasing returns to scale in the production function (a consequence of Jensen's inequality).

Higher labor mobility also raises average GDP but the effect is substantially smaller. Relative to the baseline case, the unemployment rate falls by 0.58 percentage points, GDP rises by 0.44 percentage points, and consumption rises by 0.53 percentage points. While higher labor mobility cannot replicate the optimal monetary policy regime, the gains are substantial: A drop of 35 basis points in the unemployment rate corresponds to about 1,160,000 fewer unemployed who would find a job in the euro area. Recall also that the cyclical unemployment rate in the United States is about 3%. By pushing the unemployment rate from 4.55 to 3.97%, higher labor mobility in the euro area would reduce the gap between the two currency unions by more than a quarter.

The dynamic implications of higher labor mobility are illustrated in Fig. 4, which shows the impulse responses to both a positive and a negative TFP shock. The case with baseline mobility has been discussed before. The dotted lines in the figure show corresponding reactions in a high-labor-mobility environment. While the initial unemployment response is similar in both scenarios, the half-life of an initial increase in unemployment is about 3 years as opposed to about 7 years in the baseline. This is driven by a quicker and larger net outflow of workers and goes along with a more muted response in wages.

5.2. Effects on welfare

The combination of downward nominal wage rigidity and a fixed exchange rate results in higher mean unemployment. Under an optimal exchange rate policy, an economy would be at full employment at all times. Being part of a currency union entails welfare costs that are reflected in lower average consumption levels due to inefficiencies in the labor market. The last section showed that labor mobility increases welfare in currency unions both by reducing fluctuations in unemployment and reducing mean unemployment.

Here we quantify the welfare benefits of labor mobility for countries in a currency union. For a country in state $S_{i,t} = \left\{A_{i,t}, W_{i,t-1}, \mathbb{N}^{w}_{i,t-1}, K_{i,t-1}\right\}$, we define the welfare cost of being in the currency union as the percentage increase in lifetime consumption $\lambda_{i,t}$ needed to make the household as well-off as it would be living in an economy with an optimal flexible exchange rate. We calculate this cost separately for capital owners and workers. For capital owners, $\lambda^{k}_{i,t}$ is given implicitly as the solution to

$$\mathbb{E}_t\left[\sum_{j=0}^{\infty}\beta^j\left\{U\left(c^{k}_{i,t+j}(1+\lambda^{k}_{i,t})\right)-U\left(c^{k,opt}_{i,t+j}\right)\right\}\right]=0, \tag{18}$$

where $c_{i,t}^{k,opt}$ is consumption in an economy where the central bank follows an exchange rate policy that ensures full employment, $L_{i,t} = L^S$. The compensation $\lambda_{i,t}^k$ is different in each small open economy because $S_{i,t}$ — the state of the economy — differs across countries. Countries that are currently at full employment do not need as much compensation as countries that are currently in recessions and experiencing high unemployment. Following SGU, we report the mean amount of compensation required across all the countries in the union: $\lambda^k = \sum_i \lambda_{i,t}^k$.

To calculate an equivalent expression for workers, we need to take into account that workers have the option of moving across countries. From (6) and (7), workers' lifetime utility is

$$V_{i,t} = U(c_{i,t}^w) - \frac{1}{\gamma}\ln(1 - n_{i,t}) + \beta \mathbb{E}_t V_{i,t+1}.$$

Then, the welfare cost $\lambda_{i,t}^w$ for a worker in country i at time t solves

$$\mathbb{E}_t \left[\sum_{j=0}^{\infty} \beta^j \left\{ U\left(c_{i,t+j}^w (1 + \lambda_{i,t}^w)\right) - U\left(c_{i,t+j}^{w,opt}\right) - \frac{1}{\gamma}\ln\left(\frac{1 - n_{i,t}}{1 - n_{i,t}^{opt}}\right) \right\} \right] = 0. \tag{19}$$

As before, we calculate λ^w as the average across all $\lambda_{i,t}^w$. Notice that in these calculations, we keep the migration rates, $1 - n_{i,t}$, fixed at their baseline values and do not recalculate them based on the adjusted consumption values $c_{i,t}^w(1 + \lambda_{i,t}^w)$.

In Panel C of Table 4, we report the welfare measures λ^k and λ^w for both the baseline economy and the economy with higher labor mobility. The table also reports the welfare gains for the economy as a whole, using the consumption shares of capital owners and workers in overall consumption. The cost of the currency peg is substantial; in the baseline economy the welfare cost is 4.11 percent of (non-stochastic steady-state) consumption. Higher labor mobility helps reduce the cost to 3.55 percent. Workers bear a larger welfare cost of the suboptimal monetary policy in the baseline economy (4.19 percent vs. 3.97 percent for capital owners), but they also gain slightly more from higher labor mobility, with their costs falling to 3.63 percent relative to 3.43 percent for capital owners.

The welfare metrics are almost entirely driven by first-order effects, that is, changes in average consumption as opposed to changes in the volatility of consumption. This is obviously the case for capital owners whose consumption is not directly tied to the volatility of domestic unemployment. But even for workers, higher labor mobility reduces their consumption volatility from 2.21 percent to 1.32 percent, but this accounts for only 0.01 percentage points of the overall welfare gains of 0.56 percentage points (average consumption rises by 0.54 percentage points). These small welfare gains from reducing consumption volatility are in line with the critique expressed by Lucas (1987) on the irrelevance of stabilization policy for welfare. Our model misses alternative costs from suboptimal monetary policy, such as inflation-induced relative price distortions emphasized in the New Keynesian literature (Woodford, 2003; Galí, 2008). These are evaluated by HPT in a similar model of labor mobility but without asymmetric wage rigidity and are quantitatively about 3 times smaller than those reported here.

5.3. Accounting for transition dynamics

So far, our discussion on how labor mobility reduces the welfare costs of a currency union has ignored the transition dynamics of switching from a regime with floating exchange rates to a regime with fixed exchange rates. In this section, we show that taking these transition dynamics into account reduces the costs of adopting a common currency, but labor mobility still reduces these costs by about 10 percent.

The experiment we consider is as follows: starting from an equilibrium with floating exchange rates, policymakers unexpectedly adopt a common currency. We then calculate the transition path to the new stationary equilibrium and the associated welfare costs. We use the same welfare criteria as in (18) and (19), and discount utility to the period of the policy change. We repeat this exercise for high labor mobility and calculate the resulting change in the welfare costs of adopting the common currency.

We assume that the transition takes 67 years (800 months). In the Appendix, we provide details on the algorithm used to calculate the transition path. In short, we guess a path for the average expected utility a worker gets abroad, V_t^*, and the average consumption of capital owners, $c_t^{k,*}$, along the transition, then solve for policy functions for each period, starting from the last period of the transition and solving backwards, then simulate transitions for a large number of countries and finally adjust our initial guess for V_t^* and $c_t^{k,*}$ until population movements and trade in goods balance at the union level.

Fig. 5 displays consumption in the initial stationary equilibrium under flexible exchange rates, along the transition path, and in the new stationary equilibrium under fixed exchange rates for workers (left) and capital owners (right). The blue line shows the dynamics under our baseline level of labor mobility, whereas the red line assumes high labor mobility. The values are averaged across all simulated countries and normalized to 1 in the initial stationary equilibrium.

As countries adopt the common currency, average unemployment rates rise immediately because countries can no longer use monetary policy to ease nominal wage constraints. This leads to an immediate drop in workers' consumption. As higher unemployment lowers the return to capital, investment falls and the capital stock declines. This fall in the capital stock accounts for the subsequent dynamics in consumption. For capital owners, consumption initially rises somewhat due to a fall in interest rates that reflect the lower return to capital. But over time, their consumption falls as well.

The corresponding welfare measures — reported in Panel D of Table 4 — reflect that consumption takes time to adjust to the new policy environment. The welfare costs of the currency union are consequently somewhat smaller compared to a simple comparison across stationary equilibria (3.69 vs. 4.11). The benefits from higher labor mobility are somewhat smaller, but are still economically meaningful as they reduce the welfare costs from 3.69 percent of consumption to 3.33 percent.

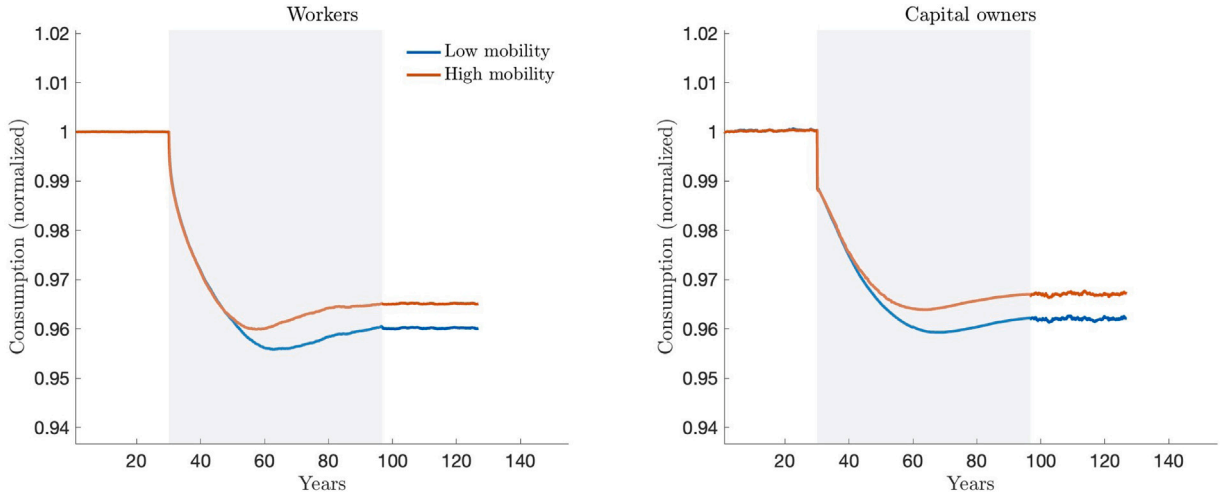


Fig. 5. Transition Dynamics from Floating to Fixed Exchange Rates.
Note: The figure plots simulated data on consumption in the initial stationary equilibrium under flexible exchange rates, along the transition path and in the new stationary equilibrium under fixed exchange rates for workers (left) and capital owners (right). The shaded area displays the transition period. The blue line shows the dynamics under our baseline level of (low) labor mobility, whereas the red line assumes high labor mobility. The values are averaged across 10,000 countries and normalized to 1 in the initial stationary equilibrium. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.4. Discussion

The welfare calculations and policy implications of our model necessarily rely on several important assumptions that deserve further discussion.

We do not consider other policies that could substitute for worker mobility. While worker mobility can address regional business cycles, it might not be the least costly channel of adjustment. For instance, fiscal transfers across member states — including a progressive union-wide income tax — could also stabilize demand and thereby reduce union-wide unemployment. In a recent paper, Pavia and Proebsting (2026) show that, empirically, fiscal transfers are the dominant mechanism for stabilizing business cycles across U.S. states, especially in the short run. Such transfers would obviate the need for workers to move. Although not compatible with the principles of the Single Market, time-varying tariffs (Monacelli, 2025) or cyclical capital controls that restrict international borrowing and lending (SGU) can also address imbalances of supply and demand in the presence of rigid nominal wages and can therefore lead to significant welfare improvements through reductions in unemployment.

Additionally, it is not clear that greater wage flexibility necessarily implies welfare improvements once additional frictions are accounted for—this is particularly true in a currency union where exchange rate adjustment is impossible. Galí and Monacelli (2016) show that, in a small open economy setting, increased wage flexibility may reduce welfare if the country is maintaining a fixed exchange rate and nominal prices are rigid. Since we assume prices to be flexible, relaxing the wage constraint is generally welfare-improving in our context.

Finally, while our paper focuses on business cycle shocks that generate persistence in GDP, consumption, investment, and unemployment as observed in the data, labor mobility likely plays a more important role at somewhat longer horizons for particularly persistent shocks. Medium-run structural transformations caused by changing patterns in international comparative advantage (e.g. the rise of China in global markets), technological progress (e.g. the effect of automation on the manufacturing sector), substantial changes in governments' fiscal policy (e.g. in the wake of the European debt crisis) or the idiosyncratic fate of specific industries or large firms (e.g. the impact of Nokia's decline on the Finnish economy) can put currency unions under pressure if these shocks are asymmetric across member states and nominal wages are slow to adjust. In the United States, labor mobility is empirically a key adjustment mechanism, absorbing about two-thirds of shocks in states' employment at a horizon of five years (Foschi et al., 2025). Our model supports the relevance of this mechanism by showing that unemployment rates in the euro area would return to normal after a recession at twice the speed if labor mobility were at U.S. levels.

5.5. Labor mobility in and out of a fiscal union

The results in the previous section show that labor mobility partially offsets the costs of being in a currency union. Changes in the local population, however, can also pose a challenge to fiscal solvency. Tax revenues tend to fall during recessions as economic activity declines. Out-migration amplifies these revenue losses by eroding the tax base during recessions. In contrast, during expansions, tax revenues rise, and the predictable inflow of migrants looking for work raises revenues even more.

Table 5
Role of the Fiscal Union.

	Fiscal Union Taxes on capital owners		National Budget Constraint Taxes on labor	
	Low mobility	High mobility	Low mobility	High mobility
SD GDP (%)	3.71	4.10	3.71	4.21
SD Net migr. (%)	0.23	0.55	0.25	0.61
Reg. net migr. on unempl.: β	−0.08	−0.26	−0.09	−0.33
Mean cyclical unempl. (%)	4.55	3.97	4.46	3.99
SD unempl. (%)	2.62	1.80	2.55	1.72
SD tax rate (%)	0.00	0.00	0.46	0.52
SD tax revenue (%)	3.71	4.10	3.25	3.70
Welfare costs of currency union				
Worker	4.19	3.63	4.25	3.75
Capital owner	3.97	3.43	3.63	3.26
Average	4.11	3.55	4.02	3.56

Notes: Fiscal union corresponds to $\theta = 0$; national budget constraint corresponds to $\theta = 0.2$. Welfare costs are expressed in consumption-equivalent units (in percent).

In this section, we briefly consider the consequences of labor migration under the assumption that national governments have to make up some of the lost revenue by adjusting their tax policy. In our baseline model, the labor income tax was constant. Here, we consider the consequences of requiring some adjustment in tax rates to help offset the decline in revenue. We maintain the assumption of a balanced budget and abstract from government spending (i.e., $G = 0$). We model the labor tax as adjusting according to the following rule:

$$\tau_{i,t}^{w} = \bar{\tau}_{i}^{w} - \theta \times \frac{\mathbb{N}_{i}^{k}\bar{T}_{i}^{k} + \bar{\tau}_{i}^{w}\mathbb{N}_{i,t-1}^{w}W_{i,t}L_{i,t}}{\mathbb{N}_{i,t-1}^{w}W_{i,t}L_{i,t}},$$

where $\bar{\tau}_{i}^{w}$ and $\bar{T}_{i}^{k}$ denote the steady-state labor tax rate and steady-state transfer to capital owners, respectively. The parameter $\theta \in [0, 1]$ governs the extent to which tax rates adjust in response to falling revenue.

This rule nests two polar cases. In our baseline specification, $\theta = 0$, the labor tax stays constant, and all fiscal adjustment occurs through changes in transfers to capital owners. Because capital owners have access to financial markets, our baseline case is effectively a fiscal union. In a fiscal union, temporary swings in tax revenues are insured by other countries and, as a result, workers are insulated from the impact of local shocks on the government budget constraint. For countries that are not part of a fiscal union, temporary budget shortfalls must be met by tax policies that raise revenue. In the extreme case, $\theta = 1$, the labor tax fully adjusts each period to ensure a constant level of labor tax revenue. The shift away from a fiscal union, toward requiring tax changes on workers, introduces a feedback mechanism between local market conditions, the government budget constraint, and the decision to migrate. Negative shocks induce workers to relocate to another country, reducing the local tax base and lowering revenue. The contraction in the tax base requires an increase in the labor tax rate, which further increases their incentive to migrate. Conversely, a positive shock provides an incentive for net in-migration. This generates additional tax revenue and allows the government to lower tax rates on labor income, further encouraging workers to enter.

Table 5 illustrates the effects of endogenous tax adjustment on migration by comparing our baseline model ($\theta = 0$) with an intermediate case where some of the tax adjustment is shifted to workers ($\theta = 0.2$). We report an abbreviated list of moments to focus on the variables most affected by the fiscal union and changes in taxation. Starting with the first two columns, we see the results familiar from Table 3. When fluctuations in the government's revenue requirement are borne fully by capital owners and capital owners can insure risk, increased labor mobility increases the volatility of net migration and aggregate GDP. Mean unemployment declines from 4.55 percent to 3.97 percent, and the standard deviation of unemployment falls from 2.62 percent to 1.80 percent. Increased labor mobility increases the volatility of GDP, which increases the volatility of tax revenue. These revenues are rebated back to capital owners and the average tax rate on labor remains at zero. The welfare costs of the currency union fall with labor mobility for both workers and for capital owners.

Columns 3 and 4 show the effects of shifting the risk associated with balancing the government's budget onto workers. Net migration is now more responsive to changes in unemployment — the benchmark net migration slope coefficient is −0.26 and increases to −0.33 — reflecting the amplification effect of changes in the tax rate. Increased migration helps reduce the mean level of unemployment to 3.99. The spread of tax rates on labor increases from zero to 0.46 percent at the low level of labor mobility. With higher labor mobility, the government must adjust the tax rate even more (a spread of 0.52 percent). The presence of downwardly rigid wages means that the change in the tax rate is not symmetric across expansions and contractions. Workers have a higher cost of being in the currency union in the ($\theta = 0.2$) scenario. Note that capital owners are better off in the ($\theta = 0.2$) scenario than they were in the fiscal union, even with close to full insurance of their tax liabilities. Because they can insure most of the risk, capital owners benefit from the amplification effect on labor mobility, which lowers unemployment and relaxes the constraint on wages.

6. Conclusion

Unemployment rates tend to be higher and more volatile in the euro area than in the United States. While there are several differences between the two regions, one key difference is the greater degree of labor mobility in United States. For a given increase

in the unemployment rate, workers are about three times more likely to relocate to another state (country) in the United States than in the euro area (House et al., 2026; Foschi et al., 2023). This paper deploys a plucking model of unemployment to study the impact of labor mobility on unemployment. Our analytical model incorporates frictions in the labor market as well as an endogenous migration decision. Firms are unable to freely reduce wages during economic contractions. This rigidity amplifies the costs of negative shocks and generates asymmetries in the dynamics of macroeconomic variables including unemployment. Workers make forward-looking decisions regarding where to work, subject to costs of relocating. This additional margin of adjustment — one that we think realistically captures a feature of labor markets in advanced economies — cuts against the friction of downward wage rigidity.

The model is calibrated to the dynamics of unemployment and net migration in a typical euro area country. The resulting model mimics the asymmetry in unemployment dynamics and the responsiveness of net migration to fluctuations in unemployment. A counterfactual of increasing the degree of labor mobility to that observed in the United States — holding all other parameters of the model fixed — would decrease both the level and the variability of unemployment in the euro area.

We believe that our framework provides a useful tool to study a broader set of related questions. The literature on optimum currency areas has identified several mechanisms through which countries can share consumption risk in the absence of independent monetary policy, including income diversification via capital markets, fiscal transfers, and credit markets. While several studies have quantified the importance of these channels using data for U.S. states and euro area countries (Asdrubali et al., 1996; Kohler et al., 2023), it remains an open question whether risk sharing not only smooths consumption but also reduces unemployment and, in turn, increases average consumption. An additional question that our framework could address is whether risk-sharing mechanisms improve or distort the allocation of resources across regions, and how this trade-off between insurance and efficiency affects aggregate productivity. We leave these extensions to future research and view our framework as a natural starting point for addressing them.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at https://doi.org/10.1016/j.jinteco.2026.104332.

Data availability

Research Data (Original data) (Mendeley Data).

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